Asset Price Dispersion, Monetary Policy and Macroprudential Regulation*†

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Abstract

Following the 2007–2009 Global Financial Crisis, sustained monetary expansion and tighter financial regulation have left financial markets thinner, less resilient, and more prone to instability. This paper develops a monetary model of decentralised financial exchange to account for these outcomes. The framework links search frictions and costly posting to the joint effects of monetary and regulatory policy on asset prices, quoting behaviour, and market stability. Increases in inflation or posting costs reduce quoting intensity, widen the distribution of executable prices, and raise the probability of trading breakdowns. The model replicates key post-crisis patterns such as wider spreads, higher execution costs, and an increased likelihood of flash-crash events, showing that the interaction between monetary and regulatory policy can unintentionally increase financial market fragility.

Keywords: Monetary policy; macroprudential regulation; asset price dispersion; search frictions; over-the-counter markets; market liquidity; flash crashes.

JEL Classification: D83; E44; E52; E58; G12; G14; G18.

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†The latest version of this paper is available at simon-mishricky.github.io/APD_Paper.pdf.

1 Introduction

One of the striking features of modern financial markets is that the policies introduced to prevent financial instability may themselves be generating new sources of fragility. In the aftermath of the 2007–2009 financial crisis, central banks—including the Federal Reserve, the European Central Bank, and the Bank of Japan—implemented historically unprecedented expansions in the money supply (see Gertler and Karadi 2011, Chadha et al. 2020, and Duc et al. 2022). At the same time, a wave of regulatory reforms, most notably Basel III and the Dodd–Frank Act, sought to curb systemic financial risk. Among these, the Volcker Rule has had a direct impact on market functioning by restricting proprietary trading and limiting the ability of intermediaries to act as principals.

A growing body of evidence indicates that these post-crisis policy shifts have altered the structure of trading and the cost of order execution. Market transactions have become more expensive, and extreme episodes of price dislocation, often described as "flash crashes", have become more frequent (see Jovanovic and Menkveld 2022, Jovanovic and Menkveld 2024, Trebbi and Xiao 2019, Golub et al. 2012, Easley et al. 2012, and Gayduk and Nadtochiy 2018). Several studies document a persistent decline in market-making activity, particularly in fixed income markets, following the implementation of these reforms. Adrian et al. (2017) survey the evidence and find that trading costs have risen across several asset classes since the financial crisis. Anderson and Stulz (2017) show that transaction costs remain elevated for large institutional orders, and Bao et al. (2018) and Dick-Nielsen and Rossi (2019) report that the cost of executing large sales increases sharply in the post-crisis period. The role of dealers as intermediaries has also diminished: Choi et al. (2024) and Bessembinder et al. (2018) find that dealers increasingly avoid acting as principal, especially when capital constraints bind.

Beyond higher trading costs, empirical evidence points to greater market fragility. Adrian et al. (2015) show that bond markets have become more prone to sudden breakdowns in intermediation capacity, while Duffie (2012) and Duffie (2018) demonstrate that post-crisis regulation raised the cost of market making and weakened the resilience of intermediaries under stress. In equity and dealer markets, changes in price posting and execution behaviour have also been observed under new regulatory constraints (Hendershott et al. 2011, Jovanovic and Menkveld 2022, Jovanovic and Menkveld 2024).

In parallel, a growing empirical literature has linked persistently loose monetary policy to broader patterns of financial instability. Grimm et al. (2023) analyse more than a century of data across advanced economies and find that extended periods of monetary accommodation are associated with a higher probability of financial crises. This effect is strongest when accommodative policy coincides with rapid credit growth. Similar evidence is provided by Boissay et al. (2023), Schularick et al. (2021), and Jiménez et al. (2023), which show that prolonged periods of low rates amplify systemic risk and lead to more volatile asset valuations. These studies collectively suggest that the joint stance of monetary and regulatory policy has profound and sometimes unintended effects on market stability.

Existing theoretical models provide only partial explanations for these developments. Standard inventory-cost models such as Grossman and Miller (1988) explain persistent execution costs through dealers' risk aversion and capital constraints but remain static and cannot reproduce abrupt breakdowns in trading activity. Search-based models of over-the-counter trade, beginning with Duffie et al. (2005) and extended by Lagos and Rocheteau (2009), incorporate frictions in matching and bargaining that generate liquidity premia, yet they abstract from the strategic pricing behaviour of intermediaries. More recent frameworks such as Jovanovic and Menkveld (2022) introduce costly participation and random entry, capturing the incentive for market participants to refrain from quoting, but they typically focus on one side of the market and exclude the role of monetary conditions. As a result, the literature does not fully explain why trading activity and price dispersion respond jointly to monetary expansion and regulatory costs, nor why flash-crash-like events can occur endogenously even in periods of apparent stability.

This paper develops a monetary model of decentralised financial exchange that integrates these missing elements into a single framework. The model shows how the stance of monetary policy and the cost of broker participation jointly determine the structure of bid and ask price distributions, the frequency of trading, and the occurrence of price collapses. Investors trade a real asset using fiat money through brokers who post prices under costly entry and random search, following the framework of Burdett and Judd (1983). The real quantity of money influences the number of active quotes through its role as a medium of exchange, while posting costs capture regulatory or balance-sheet constraints. Their interaction determines the intensity of quote competition, the spread between buying and selling prices, and the likelihood of periods with no active trading.

Specifically, the model links the quantity of money and the structure of market microstructure to fundamental dimensions of trading behaviour. In particular, it shows that (*i*) the structure of bid and ask price distributions is jointly determined by monetary conditions and the cost of broker participation, (*ii*) standard measures of market performance, such as execution costs and liquidity, depend systematically on these factors, (*iii*) the propensity for market instability, including flash-crash-like events, emerges endogenously and responds to changes in policy, and (*iv*) speculative premia and discounts arise as equilibrium outcomes when investors face these trading conditions.

This paper contributes to four main strands of the literature: search-theoretic models of money, search-based models of financial trade in over-the-counter (OTC) markets, resale-option theories of asset pricing, and costly bidder participation models. By combining insights from each of these areas, we provide a unified framework that captures how monetary conditions and market frictions interact to shape asset prices, price distributions, and market stability.

From a methodological standpoint, the paper helps bridge the gap between two bodies of work that have traditionally been developed in parallel: the search-theoretic literature on monetary economics, which emphasizes macroeconomic questions such as the efficiency of exchange and the role of liquidity in general equilibrium, and the microeconomic literature on search-based financial markets, which focuses on asset trade in decentralized settings. The core structure of our model builds on the monetary framework of Lagos and Wright (2005), and in particular follows Lagos and Zhang (2020), which adapts the OTC market framework of Duffie et al. (2005) to the context of monetary exchange. Similar hybrid structures also appear in Lagos and Zhang (2019a) and Lagos and Zhang (2019b), which incorporate bilateral asset trade with monetary exchange and frictions.

Within this class of models, several papers have introduced real assets that can serve (at least partially) as media of exchange alongside money. For example, Geromichalos et al. (2007), Jacquet and Tan (2012), Lagos and Rocheteau (2009), Lester et al. (2012), and Nosal and Rocheteau (2013) explore how assets that are useful in facilitating consumption trades acquire a liquidity premium—defined as the wedge between the market price and the fundamental value implied by future dividends. When the asset is useful in exchange, its market price exceeds the expected present discounted value of its future payoffs, reflecting its role as a store of liquidity.

A related concept is the resale premium, developed in Lagos and Zhang (2019a), Lagos and Zhang (2019b), and Lagos and Zhang (2020), which analyse how the value of an asset includes not only its dividend stream but also its usefulness in resale to future investors. This speculative component arises when investors anticipate gains from trade with others which value the asset more highly in the future. Our paper builds on this resale-option logic but applies it in a model with richer microstructure.

Specifically, we integrate the structure of Lagos and Zhang (2020) with the noisy search and costly posting model of Burdett and Judd (1983), as well as its modern applications to monetary economics as in Head et al. (2012) and Burdett et al. (2015). The framework allows us to endogenise the distribution of prices in a tractable way while preserving the macro-monetary implications of the model. We adopt a special case of the Burdett–Judd framework that is analytically convenient and well-suited to search-theoretic environments, see, Wang et al. (2020) and Mortensen (2005).

Alternative approaches to generating price dispersion arise in the literature on endogenous bidder participation in auction theory. Jovanovic and Menkveld (2022) adapt the model of Levin and Smith (1994) by assuming bidders are uncertain about how many others choose to participate. This removes the multiplicity of asymmetric pure-strategy equilibria and leaves a unique symmetric mixed-strategy equilibrium. Their model generates bid price dispersion with a closed-form distribution that matches the empirical shape of order book bids, though it lacks a monetary channel and resale considerations. The framework in Jovanovic and Menkveld (2022) is closely related to earlier common-value auction models such as Hausch and Li (1993), and the all-pay auction setting in Baye et al. (1996), which also produce strategic dispersion in posted prices.

On the ask side, a parallel literature studies price distributions arising from uncertain customer arrival or limited trading opportunities. For example, Prescott (1975) analyzes a multi-unit auction in which the number of trading opportunities is uncertain, yielding an ask price distribution that is endogenously right-skewed. Similarly, models such as Shilony (1977), Rosenthal (1980), and Varian (1980) generate price dispersion by assuming that sellers face a mix of informed and uninformed (or "captive") customers—whether due to geography, switching costs, or informational frictions.

By bringing together these strands of research—monetary frictions, noisy price posting, speculative resale motives, and endogenous participation—we construct a unified framework capable of explaining both macro-level phenomena such as the impact of inflation and bidder participation on asset markets, and micro-level features such as price dispersion, flash crashes, and bid-ask spreads.

The rest of this paper is organised as follows. Section 2 describes the basic model. Equilibrium is characterised in Section 3. Section 4 presents the theoretical implications of the model for asset prices. Section 5 presents the theoretical implications of the model to financial liquidity. Section 6 provides a discussion of the results in the context of the literature. Section 7 concludes. All proofs and supplementary derivations are available from the author upon request.

2 Model

Time is represented by a sequence of periods indexed by $t = 0, 1, \ldots$ Each time period is divided into two subperiods in which different activities take place. There is a continuum of infinitely lived agents called *investors*, each identified with a point in the set $\mathcal{I} = [0, 1]$. There is also a continuum of infinitely lived agents called *asking brokers*, each identified with a point in the set $\mathcal{B}^A = [0, 1]$, and a continuum of infinitely lived agents called *bidding brokers*, each identified with a point in the set $\mathcal{B}^B = [0, 1]$.[1] All agents discount across periods with factor, $\beta \in (0, 1)$.

There is a continuum of production units with measure $A^s \in \mathbb{R}_{++}$ that are active every period. Every active unit yields an exogenous *dividend* $y_t \in \mathbb{R}_+$. Each active unit yields the same dividend, therefore $y_t A^s$ is the aggregate dividend. At the beginning of every period, each active unit is subject to an independent idiosyncratic shock that renders it permanently unproductive with probability $1 - \eta \in [0, 1)$. If a production unit remains active, its dividend in period t is $y_t = \gamma_t y_{t-1}$, where γ_t is a nonnegative random variable with cumulative distribution function Φ, i.e., $\mathbb{P}(\gamma_t \leq \gamma) = \Phi(\gamma)$, and mean $\bar{\gamma} \in (0, (\beta\eta)^{-1})$.

The time-t dividend becomes public knowledge at the beginning of period t. At that time each failed production unit is replaced by a new unit that pays y_t in the initial period and follows

[1] It is possible to model asking and bidding brokers as a single representative broker type that performs both functions. However, such an approach complicates the exposition and obscures the distinction between the two sides of the market. For clarity, we treat asking and bidding brokers as separate agent types. This separation is purely expositional and does not affect the equilibrium results.

the same process as other active units thereafter. The dividend of the initial set of production units, $y_0 \in \mathbb{R}_{++}$, is given at $t = 0$. In the second subperiod of every period, every agent has access to a linear production technology that transforms effort into a perishable homogeneous consumption good.

For each active production unit there exists a durable and perfectly divisible *equity share* that represents ownership of the production unit and entitles the holder to its dividends. At the start of every period $t \geq 1$, each investor receives an endowment of $(1-\eta)A^s$ equity shares associated with the new production units (when a production unit fails, its equity share is extinguished). There is a second financial instrument, *money*, that is intrinsically useless. The quantity of money at time t is denoted A_t^m. The initial quantity of money, $A_0^m \in \mathbb{R}_{++}$, is given, and $A_{t+1}^m = \mu A_t^m$, with $\mu \in \mathbb{R}_{++}$. A monetary authority injects or withdraws money via lump-sum transfers or taxes to investors in the second subperiod of each period. At the beginning of period $t = 0$, each investor is endowed with a portfolio of equity shares and money.[2] All financial instruments are perfectly recognisable, not forgeable, and can be traded among agents in every subperiod.

In the second subperiod of every period, all investors can trade the consumption good produced in that subperiod, equity shares, and money in a spot Walrasian market. In the first subperiod of every period, trading is organised as follows: each broker posts a nominal price p, taking as given the distribution of prices posted by all other brokers (for asking brokers, this is described by the cumulative distribution function (CDF) $A_t(p)$, and for bidding brokers, by the CDF $B_t(p)$). Investors know the distributions $A_t(p)$ and $B_t(p)$ but can only contact, and hence can only trade with, a random sample of brokers. An investor contacts $k \in \{0, 1, 2, \dots\}$ brokers with probability $\alpha_k \in (0, 1)$. Once an investor and broker have met, the transaction takes place at the price posted by the broker. Brokers act as intermediaries, executing trades on behalf of investors in the decentralised market. After the transaction is completed, the investor and broker part ways.[3] The timing assumption is that the round of trade between investors and brokers takes place in the first subperiod of a typical period t, and ends before the equity shares yield dividends.[4] Hence, equity is traded *cum-dividend* in the first subperiod, but *ex-*

[2] We assume that brokers do not hold financial assets. This assumption allows us to abstract from the broker's portfolio problem in the first subperiod, which is not essential for the questions we study in this paper. See Lagos and Zhang (2019b) or Lagos and Zhang (2020) for a treatment of the broker's portfolio problem in this class of models.

[3] See Zhang (2018) for an OTC model with long-term relationships between investors and dealers.

[4] As in previous search models of OTC markets, e.g., Duffie et al. (2005) and Lagos and Rocheteau (2009),

dividend in the Walrasian market of the second subperiod. We assume that agents cannot make binding commitments, that there is no enforcement, and that histories of actions are private, which precludes any borrowing or lending, so all trade must be *quid pro quo*. This assumption, together with the structure of preferences described below, creates the need for a medium of exchange.[5,6]

A broker's preferences are given by

$$\mathbb{E}_0^B \sum_{t=0}^{\infty} \beta^t (c_t - h_t),$$

where c_t and h_t denote a broker's consumption of the homogeneous good that is produced, traded, and consumed in the second subperiod of period t, and the utility cost of exerting h_t units of effort to produce this good.[7] The expectation operator $\mathbb{E}_0^B$ is with respect to the probability measure induced by the random trading process in the first subperiod.[8]

An investor's preferences are given by

$$\mathbb{E}_0^I \sum_{t=0}^{\infty} \beta^t (\varepsilon_i y_t + c_t - h_t),$$

where c_t is consumption of the homogeneous good that is produced, traded, and consumed in the second subperiod of period t, and h_t is the utility cost of exerting h_t units of effort to produce this good. The variable y_t is the quantity of the dividend good that an investor consumes at the end of the first subperiod of period t, and ε_i denotes the realisation of a preference shock that is distributed independently over time and across agents, with cumulative distribution function G, where $\varepsilon_i \in \{\varepsilon_L, \varepsilon_H\}$ with corresponding probabilities $\pi_i \in (0, 1)$ and $\bar{\varepsilon} = \sum_{i\in\{L,H\}} \pi_i \varepsilon_i$. An investor learns his realisation ε_i at the beginning of period t, before trading has commenced.

an investor must own the equity in order to consume the dividend flow of the consumption good in the first subperiod.

[5]Under these conditions, there cannot exist a futures market for the equity share or the dividend, so an agent who wishes to consume the dividend must hold the equity share at the time the dividend is paid. A similar assumption is typically made in search models of financial OTC trade, e.g., Duffie et al. (2005), Lagos and Rocheteau (2009), Lagos and Zhang (2019b), or Lagos and Zhang (2020).

[6]See Lagos and Zhang (2019b) for a similar model where investors can buy equity with *margin loans*.

[7]Brokers receive no utility from the dividends of the equity share, so they have no motive to purchase equity on their own account in the first subperiod. This assumption can be relaxed, but it is standard in the search-based OTC literature, e.g., Duffie et al. (2005), Lagos and Rocheteau (2009), or Weill (2007).

[8]For analytical tractability, we abstract from brokers holding equity on their own account. While in reality dealer positions contribute to observed trade volume, modelling such behaviour would add unnecessary complexity and is not essential for addressing the questions pursued in this paper. Accordingly, we do not model trade volume explicitly and instead proxy trading activity by quote intensity and execution frequency, see Lagos and Zhang (2019b).

The expectation operator $\mathbb{E}_0^I$ is with respect to the probability measure induced by the dividend process, the investor's preference shock, and the random trading process in the first subperiod of each period t.

3 Equilibrium

Consider the determination of allocations in the first subperiod of period t for an investor who enters the period with portfolio $\mathbf{a}_t \equiv (a_t^m, a_t^s)$, consisting of $a_t^m \in \mathbb{R}_+$ units of money and $a_t^s \in \mathbb{R}_+$ units of equity, and preference type ε_i.

Let $W_t(\mathbf{a}_t)$ denote the maximum expected discounted payoff of an investor holding portfolio $\mathbf{a}_t$ at the beginning of the second subperiod of period t, after dividends have been realised. Suppose the investor enters period t with (pre-trade) portfolio $\mathbf{a}_t \equiv (a_t^m, a_t^s)$ and preference type ε_i. If the investor meets a broker, then they are able to trade and their post-trade portfolio is $\bar{\mathbf{a}}_t(\mathbf{a}_t, \varepsilon_i) \equiv (\bar{a}_t^m(\mathbf{a}_t, \varepsilon_i), \bar{a}_t^s(\mathbf{a}_t, \varepsilon_i))$, and,

$$\begin{aligned}(\bar{a}_t^m(\mathbf{a}_t, \varepsilon_i), \bar{a}_t^s(\mathbf{a}_t, \varepsilon_i)) = & \underset{(\bar{a}_t^m, \bar{a}_t^s) \in \mathbb{R}_+^2}{\arg\max} \; \varepsilon_i y_t \bar{a}_t^s + W_t\left(\bar{\mathbf{a}}_t\right) \\ & \text{s.t. } \bar{a}_t^m + p\bar{a}_t^s \leq a_t^m + pa_t^s.\end{aligned} \tag{1}$$

where p is the dollar price of equity in the first subperiod of period t.

We now analyse the investor's problem in a typical period. Let $V_t(\mathbf{a}_t, \varepsilon_i)$ denote the maximum expected discounted payoff of an investor with type ε_i and portfolio $\mathbf{a}_t = (a_t^m, a_t^s)$ at the beginning of the first subperiod of period t. Let ϕ_t^m be the real price of money, and ϕ_t^s be the *ex-dividend* price of equity in the second subperiod of period t (both expressed in terms of the second subperiod consumption good). In the second subperiod, the investor chooses consumption of the homogeneous good c_t, the utility cost of production h_t, and the end-of-subperiod portfolio $\tilde{\mathbf{a}}_{t+1} \equiv (\tilde{a}_{t+1}^m, \tilde{a}_{t+1}^s)$, recognising that at the end of the second subperiod a fraction $1 - \eta$ of equity is rendered unproductive and replaced. Therefore, we have,

$$W_t(\mathbf{a}_t) = \max_{(c_t,h_t,\tilde{\mathbf{a}}_{t+1})\in\mathbb{R}^4_+} \left[c_t - h_t + \beta\mathbb{E}_t \sum_{i\in\{L,H\}} \pi_i V_{t+1}(\mathbf{a}_{t+1},\varepsilon_i) \right] \tag{2}$$

$$\text{s.t. } c_t + \boldsymbol{\phi}_t\tilde{\mathbf{a}}_{t+1} \leq h_t + \boldsymbol{\phi}_t\mathbf{a}_t + T_t$$

$$\mathbf{a}_{t+1} = \left[\tilde{a}^m_{t+1}, \eta\tilde{a}^s_{t+1} + (1-\eta)A^s\right],$$

where $\mathbb{E}_t$ is the conditional expectation over the next-period realisation of the dividend, T_t is the real value of the time-t lump-sum monetary transfer (a tax if negative), $\boldsymbol{\phi}_t \equiv (\phi^m_t, \phi^s_t)$, and $\boldsymbol{\phi}_t\mathbf{a}_t$ denotes the dot product of $\boldsymbol{\phi}_t$ and $\mathbf{a}_t$. Since ε_i is independently and identically distributed over time, $W_t(\mathbf{a}_t)$ is independent of ε_i, and the portfolio that each investor chooses to carry into period $t+1$ is independent of ε_i.

We can now write the value function of an investor who enters the first subperiod of period t with portfolio $\mathbf{a}_t$ and preference type ε_H,

$$\begin{aligned} V_t(\mathbf{a}_t,\varepsilon_H) = \alpha_0 W_t(\mathbf{a}_t) &+ \alpha_1 \int \{\varepsilon_H y_t \bar{a}^s_t(\mathbf{a}_t,\varepsilon_H) + W_t\left[\bar{\mathbf{a}}_t(\mathbf{a}_t,\varepsilon_H)\right]\} \,\mathrm{d}A_t(p) \\ &+ \alpha_2 \int \{\varepsilon_H y_t \bar{a}^s_t(\mathbf{a}_t,\varepsilon_H) + W_t\left[\bar{\mathbf{a}}_t(\mathbf{a}_t,\varepsilon_H)\right]\} \,\mathrm{d}\left\{1 - [1-A_t(p)]^2\right\} \\ &\ldots, \end{aligned} \tag{3}$$

where $A_t(p)$ is the cumulative distribution function of ask prices. This expression indicates that an investor with a high valuation ε_H, and thus a desire to purchase equity (as shown in Lemma 1 below), faces the following outcomes. With probability α_0, the investor does not observe any prices and consequently enters the second subperiod of period t with an unchanged portfolio $\mathbf{a}_t$ without making a purchase. With probability α_1, the investor observes one price drawn from the ask distribution $A_t(p)$ and purchases equity at that price. With probability α_2, the investor observes two prices and chooses to trade at the lower of the two, which follows the distribution $1 - [1 - A_t(p)]^2$. More generally, with probability α_k, the investor observes $k \geq 1$ prices and transacts at the minimum of these observed prices, which is distributed according to $1 - [1 - A_t(p)]^k$. Hence, the value function in equation (3) can be simplified and expressed as follows:

$$V_t(\mathbf{a}_t,\varepsilon_H) = \alpha_0 W_t(\mathbf{a}_t) + \int \sum_{k=1}^{\infty} \alpha_k k[1-A_t(p)]^{k-1} \{\varepsilon_H y_t \bar{a}^s_t(\mathbf{a}_t,\varepsilon_H) + W_t\left[\bar{\mathbf{a}}_t(\mathbf{a}_t,\varepsilon_H)\right]\} \,\mathrm{d}A_t(p). \tag{4}$$

Similarly, the value function for an investor who enters the first subperiod of period t with portfolio $\mathbf{a}_t$ and a low valuation type ε_L,

$$\begin{aligned} V_t(\mathbf{a}_t, \varepsilon_L) = \alpha_0 W_t(\mathbf{a}_t) &+ \alpha_1 \int \left\{ \varepsilon_L y_t \bar{a}_t^s(\mathbf{a}_t, \varepsilon_L) + W_t \left[\bar{\mathbf{a}}_t(\mathbf{a}_t, \varepsilon_L) \right] \right\} \mathrm{d}B_t(p) \\ &+ \alpha_2 \int \left\{ \varepsilon_L y_t \bar{a}_t^s(\mathbf{a}_t, \varepsilon_L) + W_t \left[\bar{\mathbf{a}}_t(\mathbf{a}_t, \varepsilon_L) \right] \right\} \mathrm{d} \left\{ [B_t(p)]^2 \right\} \\ &\dots, \end{aligned} \tag{5}$$

where $B_t(p)$ is the cumulative distribution function of bid prices. This expression indicates that an investor with a low valuation ε_L, and thus a desire to sell equity (as shown in Lemma 1 below), faces the following outcomes. With probability α_0, the investor does not observe any prices and consequently enters the second subperiod of period t with an unchanged portfolio $\mathbf{a}_t$ without making a sale. With probability α_1, the investor observes one price drawn from the bid distribution $B_t(p)$ and sells equity at that price. With probability α_2, the investor observes two prices and chooses to trade at the higher of the two, which follows the distribution $[B_t(p)]^2$. More generally, with probability α_k, the investor observes $k \geq 1$ prices and transacts at the maximum of these observed prices, which is distributed according to $[B_t(p)]^k$. Hence, the value function in equation (4) can be simplified and expressed as follows:

$$V_t(\mathbf{a}_t, \varepsilon_L) = \alpha_0 W_t(\mathbf{a}_t) + \int \sum_{k=1}^{\infty} \alpha_k k [B_t(p)]^{k-1} \left\{ \varepsilon_L y_t \bar{a}_t^s(\mathbf{a}_t, \varepsilon_L) + W_t \left[\bar{\mathbf{a}}_t(\mathbf{a}_t, \varepsilon_L) \right] \right\} \mathrm{d}B_t(p). \tag{6}$$

Summarising, the value functions (4) and (6) can be written as one expression which corresponds to the value function in (2),

$$V_t(\mathbf{a}_t, \varepsilon_i) = \alpha_0 W_t(\mathbf{a}_t) + \int \sum_{k=1}^{\infty} \alpha_k k \left\{ \varepsilon_i y_t \bar{a}_t^s(\mathbf{a}_t, \varepsilon_i) + W_t \left[\bar{\mathbf{a}}_t(\mathbf{a}_t, \varepsilon_i) \right] \right\} \mathrm{d}F_t^k(p, \varepsilon_i), \tag{7}$$

where $\mathrm{d}F_t^k(p, \varepsilon_i) := \mathbb{I}_{\{\varepsilon_i = \varepsilon_H\}} [1 - A_t(p)]^{k-1} \mathrm{d}A_t(p) + \mathbb{I}_{\{\varepsilon_i = \varepsilon_L\}} [B_t(p)]^{k-1} \mathrm{d}B_t(p)$ and $\mathbb{I}_{\{\cdot\}}$ is an indicator function that takes the value 1 if ε_i satisfies its condition and 0 otherwise.

Next we analyse the broker's problem in a typical period. Let $W_t^B(\varsigma_t)$ denote the maximum expected discounted payoff at the beginning of the second subperiod of period t for a broker who has earned spread ς_t. Then,

$$W_t^B(\varsigma_t) = \phi_t^m \varsigma_t + \beta \mathbb{E}_t V_{j,t+1}^B, \tag{8}$$

where $V_{j,t+1}^B$ denote the maximum expected discounted payoff of either an asking or bidding broker, i.e, $j \in \{a, b\}$, where a denotes an asking broker and b denotes a bidding broker, at the beginning of the first subperiod of period $t+1$. We can now write the value function of an asking broker who enters the first subperiod of period t,

$$V_{a,t}^B = \max_{p \in \mathcal{A}} \left\{ \left[\sum_{k=1}^{\infty} \alpha_k k \left[1 - A_t(p)\right]^{k-1} \right] W_t^B \left(\frac{p - \hat{p}}{\kappa_t} \right) \right\} + \alpha_0 W_t^B(0). \tag{9}$$

Here, κ_t denotes the cost of posting a quote, and $\hat{p}$ represents the mid-price quote around which bids and asks are symmetrically distributed.[9,10] The broker's objective is to choose the price that maximises expected profits, taking into account the probability that a contacted investor will accept the quote.[11] The term inside the brackets, $\alpha_k k[1 - A_t(p)]^{k-1}$, represents the probability that an investor who observes the broker's quote has received $k-1$ other quotes, and is summed over all k. Intuitively, this captures the probability that an investor who meets this broker has possibly observed $k-1$ competing prices higher than p, and will therefore transact at the broker's quote.

Similarly, the value function of a bidding broker who enters the first subperiod of period t is

$$V_{b,t}^B = \max_{p \in \mathcal{B}} \left\{ \left[\sum_{k=1}^{\infty} \alpha_k k \left[B_t(p)\right]^{k-1} \right] W_t^B \left(\frac{\hat{p} - p}{\kappa_t} \right) \right\} + \alpha_0 W_t^B(0). \tag{10}$$

The bidding broker's objective (10) is to choose the price that maximises expected profits, taking into account the probability that a contacted investor will accept the bid to sell equity.[12] The term inside the brackets, $\alpha_k k[B_t(p)]^{k-1}$, represents the probability that an investor who observes the broker's quote has received $k-1$ other quotes, and is summed over all k. Intuitively,

[9] The symbol $\hat{p}$ denotes the mid-price quote, defined as $\hat{p} = (\bar{p} + \underline{p})/2$, where $\bar{p}$ is the highest ask price and $\underline{p}$ is the lowest bid price.

[10] Because the same underlying probability structure applies to both asking and bidding brokers, as well as to investors, the equilibrium distributions of posted prices are symmetric around the mid-price $\hat{p}$. This assumption can be generalised to allow for asymmetries across agent types, but doing so does not lead to any substantive change in the results. For ease of exposition, we therefore maintain symmetric probabilities throughout.

[11] The asking broker's price support $\mathcal{A}$ is determined in equilibrium.

[12] The bidding broker's price support $\mathcal{B}$ is determined in equilibrium.

this captures the probability that an investor who meets this broker has possibly observed $k-1$ competing bids lower than p, and will therefore transact at the broker's quote.

Let A^m_{It} and A^s_{It} denote the quantities of money and equity shares, respectively, held by all investors at the beginning of the period t (after production units have depreciated and been replaced). That is, $A^m_{It} = \int a^m_t \mathrm{d}H_t(\mathbf{a}_t)$ and $A^s_{It} = \int a^s_t \mathrm{d}H_t(\mathbf{a}_t)$, where H_t is the cumulative distribution function over portfolios $\mathbf{a}_t = (a^m_t, a^s_t)$ held by investors at the beginning at the first subperiod of period t. Let $\tilde{A}^m_{It+1}$ and $\tilde{A}^s_{It+1}$ denote the total quantities of money and shares held by all investors at the end of period t, i.e., $\tilde{A}^h_{It+1} = \int_{\mathcal{I}} \tilde{a}^h_{it+1} \mathrm{d}i$ for $h \in \{s, m\}$ with $A_{It+1} = \tilde{A}^m_{It+1}$, and $A^s_{It+1} = \eta \tilde{A}^s_{It+1} + (1-\eta)A^s$. Let $\bar{A}^m_{It}$ and $\bar{A}^s_{It}$ denote the quantities of money and shares held after the first subperiod of period t by all investors who are able to trade in the first subperiod. For asset $h \in \{s, m\}$, $\bar{A}^h_{It} = \int \sum_{k=1}^{\infty} \bar{a}^h_t(\mathbf{a}_t, \varepsilon_i) \mathrm{d}F^k_t(p, \varepsilon_i)$. We are now ready to define equilibrium.

Definition 1. *An* equilibrium *is a sequence of prices $\{\phi^m_t, \phi^s_t\}^{\infty}_{t=0}$, distributions $\{A_t(p), B_t(p)\}^{\infty}_{t=0}$, portfolio allocations in the first subperiod $\{\bar{\mathbf{a}}_t(\cdot)\}^{\infty}_{t=0}$, and end-of-period portfolio $\{\tilde{\mathbf{a}}_{t+1}\}^{\infty}_{t=0}$, such that for all t: (i) the portfolios in the first subperiod solve (1); (ii) taking prices and first subperiod portfolios as given, the end-of-period portfolio solves (2); and (iii) prices are such that all Walrasian markets clear, i.e., $\tilde{A}^s_{It+1} = A^s$ (the end-of-period t Walrasian market for equity clears), $\bar{A}^s_{It} = A^s$ (the first subperiod market for equity clears), and $(\bar{A}^m_{It} - A^m_t)\mathbb{I}_{\{\phi^m_t > 0\}} = 0$ (the first subperiod market for money clears); and, (iv) taking distributions as given prices solve the brokers' problems (9) and (10).*

The first step toward characterising equilibrium is to find the post-trade portfolio allocations of the investor. The maximisation problem in (1) represents the portfolio problem of an investor with price p in the first subperiod of period t. The solution is summarised as follows:

Lemma 1. *Define $\varepsilon^*_t \equiv \frac{p\phi^m_t - \phi^s_t}{y_t}$ and*

$$\chi(\varepsilon^*_t, \varepsilon_i) \begin{cases} = 1 & \text{if } \varepsilon^*_t < \varepsilon_i \\ \in [0,1] & \text{if } \varepsilon^*_t = \varepsilon_i \\ = 0 & \text{if } \varepsilon^*_t > \varepsilon_i. \end{cases} \tag{11}$$

Consider an investor who enters the first subperiod of period t with portfolio $\mathbf{a}_t$ and valuation

ε_i. *The investor's post-trade portfolio,* $[\bar{a}_t^m(\mathbf{a}_t, \varepsilon_i), \bar{a}_t^s(\mathbf{a}_t, \varepsilon_i)]$, *is given by*

$$\bar{a}_t^m(\mathbf{a}_t, \varepsilon_i) = [1 - \chi(\varepsilon_t^*, \varepsilon_i)]\,(a_t^m + pa_t^s) \tag{12}$$

$$\bar{a}_t^s(\mathbf{a}_t, \varepsilon_i) = \chi(\varepsilon_t^*, \varepsilon_i)(1/p)(a_t^m + pa_t^s). \tag{13}$$

Lemma 1 provides a complete characterisation of the post-trade portfolios of investors in the first subperiod of period t.[13] The outcome depends on whether the investor's valuation ε_i is above or below a cutoff level ε_t^*. That is, investors with preference type $\varepsilon_i > \varepsilon_t^*$ use all their cash to purchase equity, whereas if $\varepsilon_i < \varepsilon_t^*$, they sell all of their equity holdings for cash.[14]

In what follows, we specialise the analysis to recursive monetary equilibria. That is, equilibria in which asset holdings are constant over time, i.e., $A_{It}^s = A_I^s$, and real asset prices are time-invariant functions of the aggregate dividend, i.e., $\phi_t^s = \phi^s y_t$, $p\phi_t^m \equiv \bar{\phi}_t^s = \bar{\phi}^s y_t = p\varphi^m y_t$, $\kappa_t = \kappa y_t$, $\phi_t^m A_{It}^m = ZA^s y_t$, where $Z \in \mathbb{R}_+$. Hence, in a recursive monetary equilibrium $\varepsilon_t^* = p\varphi^m - \phi^s \equiv \varepsilon^*$, $\phi_{t+1}^s/\phi_t^s = \gamma_{t+1}$, and $\phi_t^m/\phi_{t+1}^m = \mu/\gamma_{t+1}$. Throughout the analysis, we let $\bar{\beta} \equiv \beta\bar{\gamma}$, $\bar{\beta}_L \equiv \pi_L\bar{\beta}$, and $\bar{\beta}_H \equiv \pi_H\bar{\beta}$ and maintain the following assumption,

Assumption 1. *The number of quotes* $k \in \{0, 1, 2, \dots\}$ *received by an investor is Poisson distributed with mean* $\theta \in \mathbb{R}_+$. *That is, the probability* α_k *of receiving* k *quotes is given by*

$$\alpha_k = \frac{\theta^k e^{-\theta}}{k!}. \tag{14}$$

Following Wang et al. (2020) and Mortensen (2005), we assume that the number of quotes k observed by each investor follows a Poisson distribution with mean $\theta \in \mathbb{R}_+$.[15] This specification captures the independence of quote arrivals across investors and over time while allowing closed-form solutions for equilibrium price distributions. The parameter θ is the *quote intensity.* Given this structure, the following result characterises the equilibrium ask and bid price distributions, denoted by $A(p)$ and $B(p)$, respectively. Building on the framework of Burdett and Judd (1983), these distributions describe the optimal pricing behaviour of brokers posting quotes in

[13] The lemma is stated in a general form that accommodates both discrete and continuous specifications of the preference shock ε_i.

[14] Given ε_t^*, we can interpret $p\phi_t^m = \varepsilon_t^* y_t + \phi_t^s$ is the *cum-dividend* real value of equity to the marginal investor in period t.

[15] Since the number of quotes is Poisson with mean $\theta \in \mathbb{R}_+$, its variance also equals θ. The coefficient of variation CV_θ, defined as the standard deviation divided by the mean, is therefore $CV_\theta = 1/\sqrt{\theta}$, which strictly decreases in θ. This is intuitive since one would expect greater variability when there is less quoting activity in the market.

a decentralised market with random search and posting costs. Closed-form expressions for the equilibrium ask and bid distributions consistent with profit maximisation by all brokers are presented below.

Proposition 1. *If α_k is Poisson with mean $\theta \in \mathbb{R}_+$, then*

(i) There exists a unique ask price distribution consistent with profit maximisation by all asking brokers given by

$$A(p) = \frac{1}{\theta} \ln \left[\frac{\varphi^m (p - \hat{p})}{\kappa} \right], \tag{15}$$

with support $\mathcal{A} \equiv [\hat{p} + \kappa/\varphi^m, \bar{p}] \subset \mathbb{R}_+$ and the corresponding density is

$$a(p) = \frac{1}{\theta (p - \hat{p})}. \tag{16}$$

(ii) There exists a unique bid price distribution consistent with profit maximisation by all asking brokers given by

$$B(p) = 1 - \frac{1}{\theta} \ln \left[\frac{\varphi^m (\hat{p} - p)}{\kappa} \right], \tag{17}$$

with support $\mathcal{B} \equiv \left[\underline{p}, \hat{p} - \kappa/\varphi^m\right] \subset \mathbb{R}_+$ and the corresponding density is

$$b(p) = \frac{1}{\theta (\hat{p} - p)}, \tag{18}$$

(iii) The equilibrium quote intensity is given by

$$\theta = \ln \left(\frac{\varphi^m \Gamma}{\kappa} \right), \tag{19}$$

where $\Gamma \equiv \bar{p} - \hat{p} = \hat{p} - \underline{p}$ is the book-width.

An asking broker faces a trade-off between posting a higher price to extract a larger surplus conditional on execution and posting a lower, more competitive price to increase the probability of being selected by an investor. The same logic applies symmetrically to bidding brokers, who trade off offering a higher bid to attract sellers against a lower bid to retain surplus when a trade occurs. The equilibrium price distributions, $A(p)$ for asks and $B(p)$ for bids, represent the mixed-strategy solutions to these optimisation problems under random matching and Poisson-distributed quote arrivals. Price dispersion arises endogenously from brokers' strategic indifference across feasible prices.[16]

The supports of $A(p)$ and $B(p)$ correspond to the bounds at which brokers are indifferent between posting the highest and lowest feasible prices consistent with trade. For the ask side, the lowest possible ask $\hat{p} + \kappa/\varphi^m$ represents the price at which the broker just breaks even after paying the posting cost κ.[17] Similarly, on the bid side, the highest possible bid $\hat{p} - \kappa/\varphi^m$ represents the price at which a broker is indifferent between posting and withdrawing from the market.

Finally, the quote intensity θ links market liquidity to monetary conditions and the cost of market participation. In particular, the assumption that quote arrival probabilities are identical on both sides of the market implies a symmetric quote intensity $\theta > 0$. Under this symmetry, the lower bound of the real value of money satisfies $\varphi_L^m = \kappa/\Gamma$.[18] This symmetry in quote arrival ensures that both sides of the market exhibit analogous statistical properties, allowing the equilibrium price distributions to be characterised by the following corollary.

Corollary 1. *If α_k is Poisson with mean $\theta \in \mathbb{R}_+$, then*

(i) The ask density $a(p)$ strictly decreases in p, is convex and prices are therefore right-skewed.

(ii) The bid density $b(p)$ strictly increases in p, is convex and prices are therefore left-skewed.

The intuition behind these results follows directly from the brokers' indifference conditions across the price support. On the ask side, the strictly decreasing and convex shape of $a(p)$ implies that higher ask prices are posted less frequently, producing a right-skewed distribution. As the ask price rises, the probability that an investor will accept the quote falls disproportionately, so asking brokers must engage in progressively stronger price shading to remain indifferent across feasible prices. Conversely, on the bid side, the strictly increasing and convex shape of

[16] Jovanovic and Menkveld (2022) generate a similar dispersion of bids, but their model considers only the bid side of the market and abstracts from both monetary policy considerations and asset price determination.

[17] As shown in Proposition 1, the equilibrium quote intensity satisfies $\theta = \ln(\varphi^m\Gamma/\kappa)$. This condition can be rearranged to yield $\kappa = e^{-\theta}(\varphi^m\Gamma)$, indicating that κ equals the expected benefit when no investor meets the broker. Hence, κ can be interpreted as the effective cost of participation or posting in the decentralised market.

[18] Since $\varphi_L^m = \kappa/\Gamma$, it follows that κ must also be bounded from above. If κ were to increase beyond a certain point, φ_L^m would approach φ_H^m, the upper bound of the real value of money given below in Proposition 2, thereby violating the interiority condition required for a monetary equilibrium.

$b(p)$ implies that lower bid prices are posted less frequently, yielding a left-skewed distribution. As the bid price declines, the probability of a successful trade diminishes sharply, and bidding brokers must similarly shade prices more aggressively to preserve indifference across the support.

The properties of the ask and bid distributions extend naturally to their order-statistic counterparts, the best-ask and best-bid distributions, defined as the minimum and maximum of the k observed quotes, respectively.

Corollary 2. *If α_k is Poisson with mean $\theta \in \mathbb{R}_+$, then*

(i) There exists a unique best-ask distribution given by

$$\bar{A}(p) = \begin{cases} 1 - \frac{\kappa}{\varphi^m (p - \hat{p})} & \text{if } p \in \left[\hat{p} + \frac{\kappa}{\varphi^m}, \bar{p}\right) \\ 1 & \text{if } p = \bar{p}. \end{cases} \tag{20}$$

The corresponding best-ask density $\bar{a}(p)$ strictly decreases in p, is convex and prices are therefore right-skewed.

(ii) There exists a unique best-bid distribution given by

$$\bar{B}(p) = \begin{cases} 0 & \text{if } p = \underline{p} \\ \frac{\kappa}{\varphi^m (\hat{p} - p)} & \text{if } p \in \left(\underline{p}, \hat{p} - \frac{\kappa}{\varphi^m}\right]. \end{cases} \tag{21}$$

The corresponding best-bid density $\bar{b}(p)$ strictly increases in p, is convex and prices are therefore left-skewed.

This result formalises how the process of quote sampling and selection translates into the distributions of executable prices in equilibrium. The best-ask distribution, $\bar{A}(p)$, represents the probability that the lowest observed ask price is less than or equal to p, while the best-bid distribution, $\bar{B}(p)$, represents the probability that the highest observed bid price is less than or equal to p. Both are derived directly from the underlying quote distributions $A(p)$ and $B(p)$ under the Poisson sampling structure given by Assumption 1.

The convexity and skewness properties of $\bar{A}(p)$ and $\bar{B}(p)$ mirror those of $A(p)$ and $B(p)$, whereby the distributions are more concentrated near the most competitive quotes. Specifically, the best-ask distribution is right-skewed, indicating that most observed best asks cluster near the lower bound of the support, while the best-bid distribution is left-skewed, with most observed best bids clustering near the upper bound.

Having characterised the distributions of quotes and executable prices, we now turn to the determination of equilibrium asset prices implied by these trading frictions. The following result establishes the existence and uniqueness of a recursive monetary equilibrium in which the real (*ex-dividend*) equity price and the real value of money are jointly determined by investors' portfolio choices, brokers' optimal pricing strategies, and the underlying monetary environment.

The result below formalises this relationship by linking the real equity price ϕ^s and the real value of money φ^m to the posting cost κ, and the inflation rate μ. For the analysis that follows, it is convenient to define

$$\bar{\mu} \equiv \bar{\beta}_L \left\{ 1 + \frac{1}{1 - \bar{\beta}_H \eta} \cdot \frac{\pi_H}{1 - \pi_H} \left[\varepsilon_H \frac{\Gamma}{\kappa \bar{p}} + \bar{\beta}_H \eta \frac{\underline{\underline{p}}}{\bar{p}} \right] \right\}. \tag{22}$$

The following proposition characterises the equilibrium set of the recursive monetary equilibrium.

Proposition 2. *If α_k is Poisson with mean $\theta \in \mathbb{R}_+$, and $\bar{\beta} < \mu < \bar{\mu}$, then there exists a unique recursive monetary equilibrium where real (ex-dividend) equity price is given by*

$$\phi^s = \frac{\bar{\beta}_H \eta}{1 - \bar{\beta}_H \eta} \left\{ \varepsilon_H + \frac{\pi_L}{1 - \pi_L} \left[\varphi^m \hat{p} - \frac{\kappa \theta}{1 - e^{-\theta}} \right] \right\}, \tag{23}$$

and the real value of money $\varphi^m \in (\varphi^m_L, \varphi^m_H)$ is the unique solution to

$$\frac{\varepsilon_H + \phi^s}{1 - e^{-\theta}} \int_{\hat{p} + \frac{\kappa}{\varphi^m}}^{\bar{p}} \frac{\kappa}{p \left[\varphi^m (p - \hat{p}) \right]^2} \mathrm{d}p = \frac{\mu - \bar{\beta}_L}{\bar{\beta}_L}. \tag{24}$$

This proposition characterises the equilibrium relationship between the real value of money, the equity price, and the liquidity conditions prevailing in the decentralised exchange market. Given Poisson distributed quote arrivals with mean θ, and a monetary policy satisfying $\bar{\beta} < \mu < \bar{\mu}$, there exists a unique recursive monetary equilibrium in which the real equity price ϕ^s and the real value of money φ^m jointly satisfy the two conditions above.[19]

[19]The Appendix derives the expression for Z, given by $Z = \pi_L(1 - e^{-\theta}) \left[\pi_H \int_{\hat{p}+\kappa/\varphi^m}^{\bar{p}} \frac{\kappa}{p[\varphi^m(p-\hat{p})]^2} \mathrm{d}p \right]^{-1}$. This expression is omitted from the main text to maintain focus on the core equilibrium relationships. It can be shown that $\partial Z/\partial \mu < 0$ and $\partial Z/\partial \kappa < 0$, indicating that higher inflation or posting costs reduce the aggregate value of money balances held by investors.

The equilibrium expression for ϕ^s highlights two distinct components. The first term, $\frac{\bar{\beta}_H \eta}{1-\beta_H \eta} \varepsilon_H$, represents the fundamental value of equity, which is the discounted present value of the future dividend flow as perceived by high-valuation investors. The second term inside the braces, $\varphi^m \hat{p} - \frac{\kappa\theta}{1-e^{-\theta}}$, captures the real expected value of the best-bid distribution and therefore the resale premium of the asset. A low-valuation investor values the asset at the expected present value of dividends evaluated at the mean valuation across investor types. When they sell, however, the real asset price reflects the net benefit they obtain from trade. This benefit arises because the selling investor receives more than their own valuation: by transferring the asset to a high-valuation investor, they realise the additional value that the buyer places on the dividend stream, along with the resale premium captured by the expected best bid. In this way, the equilibrium real asset price reflects both the speculative gain from reallocating the asset toward higher-valuation investors and the liquidity premium generated by active trading in the market.

The second equilibrium condition implicitly defines the real value of money φ^m as the unique level consistent with optimal portfolio choices and monetary policy. The left-hand side captures the expected liquidity return from holding money, weighted by the probability of successful trade across the support of the ask distribution, while the right-hand side, $\frac{\mu-\bar{\beta}_L}{\beta_L}$, represents the inflation-induced opportunity cost of holding money. Equilibrium is attained when these two margins are equal, yielding an interior solution $\varphi^m \in (\varphi^m_L, \varphi^m_H)$.

4 Asset Prices

In this section, we examine the properties of the equilibrium asset prices derived in Proposition 2. In particular, we analyse how these prices respond to changes in monetary policy and to the degree of trading frictions in the first subperiod as captured cost of posting.

Assumption 2. *The elasticity of the value of money φ^m with respect to the posting cost κ is less than one, i.e. $\epsilon_{\varphi^m,\kappa} < 1$, where $\epsilon_{\varphi^m,\kappa} \equiv \dfrac{\partial \varphi^m}{\partial \kappa} \dfrac{\kappa}{\varphi^m}$.*

This assumption restricts the elasticity of the value of money with respect to posting cost to be strictly subunitary, i.e. changes in κ affect φ^m less than proportionally. It ensures that the effect of κ on φ^m is such that an increase in κ leads to a decrease in quoting intensity θ, which is consistent with what we would expect. Although an equilibrium also exists when this condition is violated, we focus on this case since it corresponds to the most natural setting and aligns with what is typically observed empirically.[20]

4.1 Inflation

Variations in the inflation rate μ influence investors' portfolio decisions by changing the relative attractiveness of holding money between trading rounds. Higher inflation causes a reduction in the value of money and a decline in real money balances. This reduction in real balances, together with the lower real value of money, leads to a fall in quote intensity θ. The decline in quote intensity reflects the fact that, under the higher inflation rate, the investor whose valuation was marginal under the lower rate is no longer indifferent between carrying cash and equity out of the first subperiod market. With fewer investors entering the decentralised market with sufficient real balances, competition among brokers weakens. As a result, the resale option available to investors becomes less valuable, and the real equity price ϕ^s falls. Naturally, the real value of money φ^m is also decreasing in the growth rate of the money supply. These arguments are formalised in Proposition 3.

Proposition 3. *If α_k is Poisson with mean $\theta \in \mathbb{R}_+$, then in the recursive monetary equilibrium:*

(i) The real value of money φ^m is decreasing with inflation μ i.e., $\partial\varphi^m/\partial\mu < 0$.

(ii) The real (ex-dividend) price of the equity share ϕ^s is decreasing with inflation μ i.e., $\partial\phi^s/\partial\mu < 0$.

Proposition 3 is particularly useful in environments where monetary policy operates primarily through adjustments in the expected inflation rate. In such settings, changes in the nominal interest rate i are associated exclusively with variations in inflation μ, rather than with changes in the real interest rate, r.[21,22]

[20] The comparative statics with respect to κ reverse direction in most of the following propositions when this condition is violated. These cases are not considered here, as violating the condition produces economically unnatural outcomes that are not typically observed.

[21] It is also possible to investigate the effect of policy changes in the real interest rate r, noting that the Fisher relation implies $1 + i = \mu/\bar{\beta}$, which is equivalent to $1 + i = (1 + r)(1 + \pi)$ with $1 + \pi \equiv \mu/\bar{\gamma}$, where π is the gross inflation rate. See Lagos and Zhang (2020) for a detailed discussion of this relationship. This analysis lies outside the scope of the present paper but can be inferred directly from the relationship above.

[22] The results of Proposition 3 are consistent with the resale–premium literature of Lagos and Zhang (2015, 2019a,b, 2020), where asset prices decline with inflation, making equity a poor hedge against inflation. This empirical relationship corresponds to what is commonly known as the 'Fed model'. For an extended discussion of the Fed model and its connection to resale-premium asset-pricing frameworks, see Lagos and Zhang (2019a).

4.2 Posting Cost

Through its effect on brokers' incentives, the posting cost κ governs how actively quotes are supplied and thus how much trading occurs in the decentralised market. Higher posting costs discourage brokers from entering, which reduces the effective quote intensity θ. The decline in quote intensity reflects the fact that, under a higher κ, some brokers who were marginal at the previous cost level are no longer indifferent between posting a quote and remaining inactive. With fewer active brokers submitting quotes, investors encounter fewer trading opportunities in each round. The reduced ability to trade raises the value of holding real money balances between trading rounds, as investors must maintain higher balances to smooth consumption and meet potential trading opportunities when they arise. Consequently, the real value of money φ^m rises with κ. This increase in the value of money dominates the decline in quote intensity, leading to a higher resale premium and therefore a higher real equity price ϕ^s.[23] In equilibrium, both the real value of money and the real price of equity increase with the posting cost, reflecting the positive relationship between intermediation frictions and the valuation of liquid assets. These mechanisms are formalised in Proposition 4.

Proposition 4. *If α_k is Poisson with mean $\theta \in \mathbb{R}_+$, then in the recursive monetary equilibrium:*

(i) The real value of money φ^m is increasing with the posting cost κ, i.e., $\partial\varphi^m/\partial\kappa > 0$.

(ii) The real (ex-dividend) price of the equity share ϕ^s is increasing with the posting cost κ i.e., $\partial\phi^s/\partial\kappa > 0$.

Proposition 4 provides a theoretical basis for interpreting regulatory interventions as effective increases in the posting cost κ. By raising the cost of intermediation, such policies reduce quote intensity and make holding real money balances more valuable, which in turn increases the real equity price ϕ^s through the resale channel.

5 Financial Liquidity

In this section, we examine how monetary policy and trading frictions jointly shape equilibrium outcomes in the recursive monetary economy. We analyse how variations in monetary

[23]Assumption 2 ensures that the effect of κ on φ^m is such that θ is decreasing in κ, as stated in Proposition 5 below. Hence, the effect of κ on φ^m does not dominate the direct effect that κ has on θ. However, this dominance holds in the case of the real equity price ϕ^s, where the increase in φ^m more than offsets the decline in quote intensity and direct effect of κ.

conditions and in the degree of market frictions influence key aspects of decentralised trade and asset pricing. Specifically, we study how these factors affect quote intensity, the stochastic dominance properties of the price distributions, the extent of price dispersion, and the determination of bid–ask spreads. We also examine how the interaction between monetary and regulatory conditions contributes to systemic vulnerabilities, including flash-crash risk, and how it shapes speculative behaviour in decentralised financial markets.

5.1 Quote Intensity

Consider first how monetary policy and posting costs affect quoting behaviour in the decentralised market. According to Propositions 2 and 3, an increase in the inflation rate μ reduces investors' incentive to maintain real balances. With fewer investors carrying adequate real balances, the value of the spread earned by brokers is reduced, which in turn lowers their incentive to post quotes. The resulting decline in quoting activity leads to a reduction in quoting intensity θ. Similarly, according to Proposition 4 and given Assumption 2, a higher posting cost κ raises the marginal cost of participation for brokers and further depresses quoting intensity θ.[24] This intuition is formalised in the following proposition.

Proposition 5. *If α_k is Poisson with mean $\theta \in \mathbb{R}_+$ and $\epsilon_{\varphi^m,\kappa} < 1$, then in the recursive monetary equilibrium, the quote intensity θ: (i) decreases with inflation μ, i.e., $\partial\theta/\partial\mu < 0$; and (ii) decreases with the posting cost κ, i.e., $\partial\theta/\partial\kappa < 0$.*

The quote intensity θ thus serves as the principal transmission channel through which monetary and regulatory policy affect outcomes in the decentralised market. As we will see, variations in θ translate policy shocks into observable changes in spreads, depth, and the overall shape of the bid and ask price distributions.

[24] Following the 2007–2009 crisis, a structural break in liquidity supply has been linked to post-crisis regulation. On July 21, 2010, the Dodd–Frank Act was signed into law. Section 619, the Volcker Rule, generally prohibits insured depository institutions and their affiliates from proprietary trading and from certain relationships with hedge funds and private equity funds, subject to exemptions and definitions. Restricting proprietary trading removed banks as deep-pocket market makers and reduced immediacy provision. The request for public comments closed on November 5, 2010, by which time market participants likely understood the substance of the forthcoming rule, helping to explain the observed break in liquidity supply near the end of 2010. The amended rule became effective on April 1, 2014, with full compliance required by July 21, 2015. The interval between these dates coincides with an accelerated decline in 50-basis-point depth and a reduction in the average number of middlemen, consistent with the view that the Volcker Rule accounted for a large part of the depth decline.

5.2 *Stochastic Dominance*

The best-ask distribution $\bar{A}(p)$ gives the probability that the lowest observed ask price is less than or equal to p, and therefore captures the behaviour of marginal sellers. When the inflation rate μ rises, the real value of money φ^m declines and investors enter the decentralised market with smaller real balances. The resulting fall in quote intensity θ reduces the number of competing ask quotes that investors observe, shifting probability mass in $\bar{A}(p)$ toward lower prices. A rise in the posting cost κ produces a similar effect. Higher κ raises the cost of participation for brokers and lowers θ, reducing the frequency of competitive quoting and shifting the distribution of executable ask prices downward. In both cases, $\bar{A}(p)$ takes smaller values for every p, indicating a downward shift of the CDF. Formally, for any p, $\bar{A}_2(p) < \bar{A}_1(p)$ when $\mu_2 > \mu_1$ or $\kappa_2 > \kappa_1$, implying that the distribution associated with lower inflation or lower posting costs first-order stochastically dominates that under higher values of either parameter. The following proposition formalises this intuition.[25]

Proposition 6. *If α_k is Poisson with mean $\theta \in \mathbb{R}_+$ and $\epsilon_{\varphi^m,\kappa} < 1$, then in the recursive monetary equilibrium the best-ask distribution $\bar{A}(p)$ satisfies: (i) if $\mu_2 > \mu_1$ then $\bar{A}_{\mu_1,\kappa}(p)$ first-order stochastically dominates $\bar{A}_{\mu_2,\kappa}(p)$; and, (ii) if $\kappa_2 > \kappa_1$ then $\bar{A}_{\mu,\kappa_1}(p)$ first-order stochastically dominates $\bar{A}_{\mu,\kappa_2}(p)$.*

The best-bid distribution $\bar{B}(p)$ gives the probability that the highest observed bid price is less than or equal to p, and therefore captures the behaviour of marginal buyers. When the inflation rate μ rises, the real value of money φ^m declines and investors enter the decentralised market with smaller real balances. The associated fall in quote intensity θ reduces the number of competing bids that investors observe, shifting probability mass in $\bar{B}(p)$ toward higher prices. A rise in the posting cost κ produces a similar effect. Higher κ raises the cost of participation for brokers and lowers θ, reducing the frequency of competitive quoting and shifting $\bar{B}(p)$ toward higher prices. Formally, for any p, $\bar{B}_2(p) > \bar{B}_1(p)$ when $\mu_2 > \mu_1$ or $\kappa_2 > \kappa_1$, implying that the distribution associated with higher inflation or higher posting costs first-order stochastically dominates that under lower values of either parameter. The following proposition formalises

[25] Proposition 6A in the Appendix shows that the ask distribution is decreasing in both μ and κ. Higher inflation reduces real balances, and higher posting costs discourage participation, both of which make high asking prices less likely in equilibrium. Only the results for the best-ask distribution are presented in the main text, as the analysis focuses on executable prices.

this intuition.[26]

Proposition 7. *If α_k is Poisson with mean $\theta \in \mathbb{R}_+$ and $\epsilon_{\varphi^m,\kappa} < 1$, then in the recursive monetary equilibrium the best-bid distribution $\bar{B}(p)$ satisfies: (i) if $\mu_2 > \mu_1$ then $\bar{B}_{\mu_2,\kappa}(p)$ first-order stochastically dominates $\bar{B}_{\mu_1,\kappa}(p)$; and, (ii) if $\kappa_2 > \kappa_1$ then $\bar{B}_{\mu,\kappa_2}(p)$ first-order stochastically dominates $\bar{B}_{\mu,\kappa_1}(p)$.*

5.3 Price Dispersion

The mean and variance of the best-ask distribution are given respectively by

$$\mathcal{M}_a = \hat{p} + \frac{\kappa\theta}{\varphi^m(1-e^{-\theta})} \quad \text{and} \quad \mathcal{V}_a = \left(\frac{\kappa}{\varphi^m}\right)^2 \left[e^{\theta} - \left(\frac{\theta}{1-e^{-\theta}}\right)^2\right]$$

These expressions summarise how the level and dispersion of executable ask prices vary with monetary and posting conditions. When the inflation rate μ increases, the real value of money φ^m declines and investors enter the decentralised market with smaller real balances. The resulting fall in quote intensity θ reduces the number of competing ask quotes that investors observe, leading to a higher average execution price and greater variation across observed asks. Consequently, both the mean $\mathcal{M}_a$ and the variance $\mathcal{V}_a$ rise with inflation. A similar mechanism operates when the posting cost κ increases. Higher κ discourages brokers from posting, lowers θ, and widens the distance between the most and least competitive ask quotes. As a result, $\mathcal{M}_a$ rises since the average executed ask price increases, and $\mathcal{V}_a$ rises as the distribution becomes more dispersed. Formally, both $\mathcal{M}_a$ and $\mathcal{V}_a$ are increasing in μ and κ, indicating that higher inflation or posting costs shift the best-ask distribution upward and amplify its cross-sectional spread. The following proposition formalises this intuition.

Proposition 8. *If α_k is Poisson with mean $\theta \in \mathbb{R}_+$ and $\epsilon_{\varphi^m,\kappa} < 1$, then in the recursive monetary equilibrium, (i) the mean $\mathcal{M}_a$ of the best-ask distribution is increasing in both μ and κ; and, (ii) the variance $\mathcal{V}_a$ of the best-ask distribution is increasing in both μ and κ.*

[26] Proposition 7A in the Appendix shows that the bid distribution is increasing in both μ and κ. Rising inflation lowers real balances and higher posting costs reduce quoting intensity, both of which shift bids downward and make lower bid prices more probable. Only the results for the best-bid distribution are presented in the main text, as the analysis focuses on executable prices.

The mean and variance of the best-bid distribution are given respectively by

$$\mathcal{M}_b = \hat{p} - \frac{\kappa\theta}{\varphi^m(1-e^{-\theta})} \quad \text{and} \quad \mathcal{V}_b = \left(\frac{\kappa}{\varphi^m}\right)^2 \left[e^\theta - \left(\frac{\theta}{1-e^{-\theta}}\right)^2\right]$$

When the inflation rate μ rises, the real value of money φ^m declines, and investors enter the decentralised market with smaller real balances. The resulting fall in quote intensity θ reduces the number of competing bids that investors observe, leading to a lower average execution price. The reduction in competition among brokers also increases the spread between high and low bids, raising the variance $\mathcal{V}_b$. A rise in the posting cost κ produces the same comparative statics. Higher κ discourages brokers from quoting, lowers θ, and widens the range of feasible bid prices. As a result, the mean $\mathcal{M}_b$ declines while the variance $\mathcal{V}_b$ increases. Formally, $\mathcal{M}_b$ is decreasing and $\mathcal{V}_b$ is increasing in both μ and κ, indicating that higher inflation or posting costs shift the best-bid distribution downward and make it more dispersed. The following proposition formalises this intuition.

Proposition 9. *If α_k is Poisson with mean $\theta \in \mathbb{R}_+$ and $\epsilon_{\varphi^m,\kappa} < 1$, then in the recursive monetary equilibrium, (i) the mean $\mathcal{M}_b$ of the best-bid distribution is decreasing in both μ and κ; and, (ii) the variance $\mathcal{V}_b$ of the best-bid distribution is increasing in both μ and κ.*

Beyond the mean and variance, a more informative way to assess dispersion is through the coefficient of variation CV. Defined as the ratio of the standard deviation to the mean, $CV_j = \sqrt{\mathcal{V}_j}/\mathcal{M}_j$ for $j \in \{a, b\}$. This statistic provides a scale-free measure of variability that captures how dispersed a distribution is relative to its average level. It serves as an important indicator of uncertainty in decentralised markets, reflecting how volatile or concentrated quotes are around their mean price. A higher coefficient of variation indicates greater heterogeneity in observed quotes and lower predictability in execution prices.

In the context of the best-bid distribution $\bar{B}(p)$, a rise in the inflation rate μ reduces the real value of money φ^m and lowers investors' real balances. The resulting decline in quote intensity θ decreases competition among brokers, leading to a fall in the mean $\mathcal{M}_b$ and a rise in the variance $\mathcal{V}_b$. Because these effects move in opposite directions, the ratio $\sqrt{\mathcal{V}_b}/\mathcal{M}_b$ increases, indicating that bids become more dispersed relative to their average level. An increase in the posting cost κ produces the same comparative statics: higher costs discourage quoting, reduce θ, and widen the spread between high and low bids. Consequently, this intuition is summarised by the following corollary.

Corollary 3. *If α_k is Poisson with mean $\theta \in \mathbb{R}_+$ and $\epsilon_{\varphi^m,\kappa} < 1$, then in the recursive monetary equilibrium, the coefficient of variation CV_b of the best-bid distribution: (i) increases with inflation, i.e., $\partial CV_b/\partial\mu > 0$; and (ii) increases with the posting cost, i.e., $\partial CV_b/\partial\kappa > 0$.*

By contrast, the coefficient of variation CV_a of the best-ask distribution $\bar{A}(p)$ exhibits no general direction with respect to these parameters. Changes in μ or κ affect the mean and variance of the best-ask distribution $\bar{A}(p)$ in offsetting ways, making the overall response of $\sqrt{\mathcal{V}_a}/\mathcal{M}_a$ ambiguous in equilibrium.

Propositions 8 and 9, together with Corollary 3, show that monetary and intermediation conditions jointly determine the degree of price dispersion in decentralised markets. These results are consistent with the patterns established in Propositions 6 and 7, but provide a more detailed characterisation of how changes in inflation and posting costs translate into differences in the distribution of executable prices and the extent of dispersion across trades.

5.4 Spreads

Proposition 8 shows that the mean price $\mathcal{M}_a$ with respect to the best-ask distribution $\bar{A}(p)$ captures the average price at which the marginal seller is willing to transact, conditional on being the most competitive offer at any given time. Proposition 9 shows that the mean price $\mathcal{M}_b$ with respect to the best-bid distribution $\bar{B}(p)$ captures the average price at which the marginal buyer is willing to transact, conditional on being the most competitive bid. Hence, the *nominal bid–ask spread* is $\mathcal{S}^m = |\mathcal{M}_a - \mathcal{M}_b|$ which is given by

$$\mathcal{S}^m = \frac{2\kappa\theta}{\varphi^m(1 - e^{-\theta})}$$

When the inflation rate μ increases, the real value of money φ^m declines, and investors enter the decentralised market with smaller real balances. The reduction in purchasing power lowers the quoting intensity θ, leading to fewer competing quotes and transactions over a wider range of offers. As a result, the nominal bid–ask spread $\mathcal{S}^m$ increases with inflation. Similarly, when the posting cost κ rises, brokers quote less and require greater expected compensation to recover their costs. The higher cost of participation directly widens the price differential between the average best-bid and best-ask quotes, while the fall in θ further amplifies this effect by reducing market competition.

Proposition 10. *If α_k is Poisson with mean $\theta \in \mathbb{R}_+$ and $\epsilon_{\varphi^m,\kappa} < 1$, then in the recursive monetary equilibrium, the nominal bid-ask spread $\mathcal{S}^m$: (i) increases with inflation, i.e., $\partial \mathcal{S}^m / \partial \mu > 0$; and (ii) increases with the posting cost, i.e., $\partial \mathcal{S}^m / \partial \kappa > 0$.*

To evaluate how these differences translate into real economic costs, it is useful to consider the *real bid–ask spread*, defined as the nominal spread scaled by the real value of money φ^m, i.e., $\mathcal{S} \equiv \varphi^m \mathcal{S}^m$. The real spread measures the effective trading cost in terms of consumption goods and therefore captures the true resource cost of market participation. In equilibrium, variations in inflation and posting costs influence the real spread only through their effects on the quoting intensity θ, linking monetary and intermediation frictions to the efficiency of asset exchange. The following result is a corollary of Proposition 10.[27]

Corollary 4. *If α_k is Poisson with mean $\theta \in \mathbb{R}_+$ and $\epsilon_{\varphi^m,\kappa} < 1$, then in the recursive monetary equilibrium, the real bid-ask spread $\mathcal{S}$: (i) decreases with inflation, i.e., $\partial \mathcal{S} / \partial \mu < 0$; and (ii) increases with the posting cost, i.e., $\partial \mathcal{S} / \partial \kappa > 0$.*

This result is novel within the monetary resale literature, as the behaviour of the real bid–ask spread in response to monetary policy and trading frictions is generally ambiguous in earlier frameworks such as Lagos and Zhang (2015). The present framework yields unambiguous comparative statics, showing that the real bid–ask spread decreases with inflation and increases with the posting cost, consistent with the empirical literature.

5.5 Flash Crash Risk

A *flash crash* is defined as the event in which no quotes are observed by investors, so that no transactions take place in the decentralised market. Since α_0 denotes the proportion of investors who receive no quote in equilibrium, it corresponds directly to the probability of a flash crash. Given that the number of quotes k received by each investor is Poisson distributed with mean θ, the probability that no quotes are observed is $\mathcal{F} = e^{-\theta}$. Following Jovanovic and Menkveld

[27] The functional form of the real bid–ask spread used here is identical to that in Jovanovic and Menkveld (2022), which yields similar comparative statics with respect to the posting cost κ. In both settings, higher intermediation costs raise the equilibrium real bid–ask spread through its direct effect and by lowering quoting intensity. Their model, however, focuses solely on the bid side of pricing and abstracts from monetary conditions and asset valuation, which are central to the present analysis.

(2022), this formulation captures the idea that flash crashes arise endogenously as a result of strategic inaction: when the expected gains from trade fall below the cost of posting, brokers optimally choose not to quote, leading to episodes of market illiquidity even in the absence of exogenous shocks or informational frictions.[28]

Monetary policy affects flash crash risk through its influence on quote intensity. Higher inflation increases the opportunity cost of holding money between trading rounds, reducing the real balances investors bring into the decentralised market. With fewer investors carrying liquidity, brokers face weaker trading opportunities and lower expected gains from posting quotes. This decline in activity compresses the value of the spread, further reducing the incentive to quote. As a result, both investor participation and broker activity fall, lowering the equilibrium quote intensity θ and increasing $\mathcal{F}$. Similarly, higher posting costs κ raise the marginal cost of quoting for brokers. When these costs become more significant relative to the expected spread, fewer brokers find it profitable to post quotes, and those that do participate less actively. The resulting decline in quote intensity again raises the probability that no quotes are observed.

In equilibrium, both tighter monetary conditions and higher trading frictions weaken market participation and increase flash crash risk, making the decentralised market more fragile. The risk of a flash crash therefore increases when inflation is high or the cost of posting rises, as summarised in the following proposition.[29]

[28] Define the *half-spread* $\mathcal{H}$ as the difference between the mid-price and the expected best-ask price or the expected best-bid price: $\mathcal{H} \equiv \mathcal{M}_a - \hat{p} = \hat{p} - \mathcal{M}_b = \kappa\theta \big/ \big(\varphi^m(1-e^{-\theta})\big) = \mathcal{S}^m/2$. Given this we can write the variance of the best-ask distribution and the best-bid distribution as $\mathcal{V} \equiv \mathcal{V}_a = \mathcal{V}_b = \kappa^2 \big/ \big((\varphi^m)^2 e^{-\theta}\big) - \mathcal{H}^2 = \kappa^2 \big/ \big((\varphi^m)^2 \mathcal{F}\big) - \mathcal{H}^2$. It is also important to note that we define the *mean-squared error* (MSE) in the following way, $\text{MSE}_a = \mathbb{E}_{\bar{A}}[(p-\hat{p})^2] = \mathcal{V}_a + (\mathcal{M}_a - \hat{p})^2 = \mathcal{V} + \mathcal{H}^2 = \text{MSE}_b \equiv \text{MSE}$, and therefore $\text{MSE} = \kappa^2 \big/ \big((\varphi^m)^2 \mathcal{F}\big)$. With $\theta = \ln(\varphi^m \Gamma/\kappa)$, this specification links the variance $\mathcal{V}$, the half-spread $\mathcal{H}$, the book width Γ, the non-crash pricing error MSE, and the flash-crash probability $\mathcal{F}$ in a single set of identities: $\text{MSE} = \mathcal{V} + \mathcal{H}^2 = \kappa\,\Gamma/\varphi^m$ and $\mathcal{F} = \kappa/(\varphi^m\Gamma)$. The inverse proportionality then has a precise meaning: MSE and $\mathcal{F}$ are reciprocal transforms of the same book-width object Γ (up to the fixed scale factor κ/φ^m). Put differently, increases in Γ raise MSE and lower $\mathcal{F}$, and the product $\text{MSE} \cdot \mathcal{F} = \kappa^2/(\varphi^m)^2$ is conserved. This makes the frequency–magnitude trade-off exact: doubling $\mathcal{F}$ halves MSE, and as $\mathcal{F} \downarrow 0$ we must have $\text{MSE} \uparrow \infty$. It also yields a testable and calibratable restriction (near-constancy of $\text{MSE} \cdot \mathcal{F}$) and a clear diagnostic: observed $(\mathcal{V}, \mathcal{H}^2)$ must add to $\text{MSE} = \kappa\,\Gamma/\varphi^m$, while any violation of $\text{MSE} \cdot \mathcal{F} = \kappa^2/(\varphi^m)^2$ points to misspecification or measurement error. A decline in MSE can reflect *more* probability mass in crash states (higher $\mathcal{F}$), not necessarily safer day-to-day conditions: by $\text{MSE} \cdot \mathcal{F} = \kappa^2/(\varphi^m)^2$, lower MSE must be offset by higher $\mathcal{F}$ (or a change in κ/φ^m). Hence low day-to-day noise in quotes, i.e., low non-crash pricing error (MSE small), can go hand-in-hand with elevated crash probability ($\mathcal{F}$ high). Thus warning that smaller MSE may reflect a reallocation of risk into rarer crash states rather than genuinely safer conditions.

[29] Corollary 2A in the Appendix derives the Value-at-Risk (quantile) functions of the best bid and best ask distributions. It demonstrates that these functions exhibit a piecewise structure composed of continuous segments separated by discrete mass points at the price boundaries, $\bar{p}$ for asks and $\underline{p}$ for bids. These mass points represent the non-negligible probability that no quotes are posted at the extremes of the market, corresponding to episodes of illiquidity or flash crashes. The size and location of the mass points depend on the ratio $\kappa/(\varphi^m\Gamma)$, which links the cost of posting quotes, κ, the value of money, φ^m, and the spread, Γ. When φ^m increases, the continuous intervals of the quantile functions widen, indicating a higher density of posted quotes and greater market depth,

Proposition 11. *If α_k is Poisson with mean $\theta \in \mathbb{R}_+$ and $\epsilon_{\varphi^m,\kappa} < 1$, then in the recursive monetary equilibrium, the flash crash risk $\mathcal{F}$: (i) increases with inflation, i.e., $\partial\mathcal{F}/\partial\mu > 0$; and (ii) increases with the posting cost, i.e., $\partial\mathcal{F}/\partial\kappa > 0$.*

This is a notable result, as it implies that policies typically designed to mitigate financial instability, such as looser monetary conditions or higher regulatory posting costs, can in this framework amplify market fragility by increasing the probability of a flash crash. When higher money growth or stricter intermediation requirements reduce quoting activity, market depth erodes and liquidity becomes more fragile, raising the likelihood that trading halts occur endogenously. This mechanism highlights a potential policy trade-off between stability in normal times and vulnerability to sudden market breakdowns.

5.6 Speculation

It is standard to define the *fundamental value* of an asset as the expected present discounted value of its future dividend stream, $\hat{\phi}_t^s \equiv \left[\bar{\beta}\eta/(1-\bar{\beta}\eta)\right]\bar{\varepsilon}y_t \equiv \hat{\phi}^s y_t$, and to refer to any transaction value in excess of this fundamental value as a *bubble*. One could argue, however, that the relevant notion of "fundamental value" should incorporate market aggregation across heterogeneous investor valuations and reflect the influence of monetary policy and trading frictions such as the frequency of trading opportunities and the market power of intermediaries, which together determine asset prices in equilibrium. To avoid semantic ambiguity, we follow Harrison and Kreps (1978) and define the *speculative premium (or discount)* as the difference between the market price and the fundamental value, $\mathcal{P}_t \equiv \phi_t^s - \hat{\phi}_t^s$, which captures the value investors assign to the right to resell the asset in the future. Investors exhibit speculative behaviour when this resale option induces them to pay more or less for the asset than they would if obliged to hold it indefinitely.[30] According to Proposition 2, the speculative premium/discount is $\mathcal{P}_t = \mathcal{P}y_t$,

as brokers find it more profitable to supply quotes. This reduces the probability mass at the boundaries and lowers flash-crash risk. In contrast, increases in κ shrink these intervals, pushing more probability mass toward the boundaries and raising the likelihood that no quotes are posted. The contraction in quoting activity translates into thinner markets and greater fragility. The piecewise structure captured by equations of Corollary 2A highlights the nonlinear response of market stability to changes in policy and participation costs: small variations in μ or κ can shift the support of the quote distributions and alter the boundary mass points in ways that sharply amplify the probability of flash crashes.

[30] Our notion of *speculative premium* corresponds to the concept of a *speculative bubble* used in the modern literature, e.g., Barlevy (2007), Abreu and Brunnermeier (2003), Scheinkman and Xiong (2003), Scheinkman (2013), and Xiong (2013), following the *resale-option theory of bubbles* introduced by Harrison and Kreps (1978).

where

$$\mathcal{P} = \frac{\bar{\beta}_H \eta}{1 - \bar{\beta}_H \eta} \left[\varepsilon_H + \frac{\pi_L}{1 - \pi_L} \varphi^m \mathcal{M}_b \right] - \hat{\phi}^s.$$

Note that, unlike Lagos and Zhang (2015, 2019b), which generate only speculative premia, the present model can produce both speculative premia and speculative discounts. This represents a novel feature of the environment and one that is consistent with findings in the empirical literature. In particular, a speculative discount $\mathcal{P} < 0$ arises when $\varphi^m \mathcal{M}_b < \varepsilon_L + \hat{\phi}^s$; equality implies no speculative component $\mathcal{P} = 0$, and otherwise there is a speculative premium $\mathcal{P} > 0$. Economically, $\varphi^m \mathcal{M}_b$ denotes the real mean best bid, the real value investors expect to obtain when selling the asset at the average bid price in the decentralised market, or, in other words, the expected *cum-dividend* price of the equity share. This value declines as inflation reduces real balances and lower quote intensity limits trading opportunities. The term $\varepsilon_L + \hat{\phi}^s$ represents the fundamental benchmark, combining the valuation of the low-type investor with the expected discounted value of future dividends.

Since Proposition 3 shows that a higher inflation rate lowers the real asset price, it follows that the speculative premium decreases with inflation. Higher inflation raises the opportunity cost of holding money and reduces the real balances investors carry into the decentralised market. With less money available, quote intensity falls and the resale option embedded in the asset loses value. Investors are therefore less willing to pay above the fundamental value, and the speculative premium declines.

In contrast, an increase in the posting cost κ raises the speculative premium. While higher posting costs reduce brokers' quoting activity and increase trading frictions, they also raise the real value of money φ^m which induces investors to hold larger real balances between trading rounds. This increase in φ^m dominates the opposing effects of κ on market participation, supporting higher asset valuations and amplifying the resale option value embedded in prices, consistent with Proposition 4.

In summary, monetary loosening reduces speculative behaviour by weakening resale opportunities, whereas higher posting costs increase it, since the decline in quote intensity is more than offset by the rise in money holdings generated by the higher real value of money that accompanies an increase in κ. These relationships are formalised in the following proposition.

Proposition 12. *If α_k is Poisson with mean $\theta \in \mathbb{R}_+$, then in the recursive monetary equi-*

librium, the speculative premium $\mathcal{P}$: (i) decreases with inflation, i.e., $\partial\mathcal{P}/\partial\mu < 0$; and (ii) increases with the posting cost, i.e., $\partial\mathcal{P}/\partial\kappa > 0$.

A novel implication of the model is that speculative premia/discounts increase in response to higher intermediation costs, even though such costs reduce quote intensity. Evidence consistent with this mechanism may already be present in the empirical literature (see Duffie 2012, 2018).

6 Discussion

This section highlights where the model's predictions line up with established stylized facts regarding the data. First, consider the shape of buy and sell quotes. Empirical work shows that buy and sell quote distributions are asymmetric and convex: bids cluster below the midprice and asks above it. This pattern is persistent across countries and asset classes, as documented in Biais et al. (1995); Goldstein and Kavajecz (2004); Hollifield et al. (2006); Næs and Skjeltorp (2006); Degryse et al. (2015). Proposition 1 generates the same shape on both sides in the model. Strategic posting with participation costs and random matching yields ask and bid distributions that are convex and skewed. Corollaries 1 and 2 correspond to the best quote and tail behaviour reported in Zovko and Farmer (2002), Weber and Rosenow (2005), and Gu et al. (2008), where best quotes concentrate near the inside and executions display long tails.

Second, broker participation has declined following post-crisis regulatory reforms. Jovanovic and Menkveld (2022) document that the number of active intermediaries in U.S. equity markets fell by more than half over an eleven-year period, with a sharp break following the introduction of the Volcker Rule in 2013–2014. Bessembinder et al. (2018) and Choi et al. (2024) show similar changes, with dealers stepping back from principal activity and quoting becoming noticeably sparser as capital and balance-sheet constraints tightened. Proposition 5 generates this reduction in broker participation in the model. Increased regulatory requirements correspond to a higher effective posting cost, which lowers broker entry and reduces quote intensity. This results in sparser quoting, consistent with the empirical evidence.

Third, declines in broker participation have been accompanied by reductions in market depth and increases in effective trading costs. Empirical studies have shown that although quoted spreads narrowed modestly in the decade following the crisis, deeper execution metrics reveal a substantial deterioration in market depth. Adrian et al. (2017) find that while bid–ask spreads in the sample of Jovanovic and Menkveld (2022) fell by around 6 percent per year, depth near the best quote fell by almost 20 percent annually, and depth within 50 basis points of the midprice declined by more than 25 percent. Quoted spreads appeared narrow, but executing large trades became more costly. Propositions 6 and 7 account for the decline in depth: when broker participation falls, quote intensity declines and fewer quotes are executed close to the midprice. Corollary 4 explains the narrowing of the quoted spread: an increase in the money growth rate can compress the bid–ask spread even as participation falls and even as regulatory costs increase. These observations show that commonly used measures of liquidity, such as the bid–ask spread, are not sufficient to assess market conditions.

Fourth, markets occasionally experience extreme dislocations, such as flash crashes. Jovanovic and Menkveld (2022) show that while the probability of observing no quotes on a given day is small, over longer horizons it implies roughly a 31 percent chance of at least one such event in their sample. Menkveld and Yueshen (2019) documents that during the May 6, 2010 Flash Crash, prices in several U.S. equities temporarily collapsed to near zero before recovering within minutes, consistent with a brief absence of liquidity rather than a change in fundamentals. Proposition 11 places this observation in the context of variation in quote intensity: when quote intensity is low, quote arrivals become sparse and the probability of periods with no posted quotes increases. The model thus explains how seemingly stable markets can become vulnerable to sudden collapses in trading activity as an endogenous consequence of participation frictions.

Fifth, speculative episodes often feature both high valuations and high trading activity. Scheinkman and Xiong (2003) and Scheinkman (2013) emphasise that speculative premia are associated with strong resale motives and elevated trade volume. Together, Proposition 5, Corollary 3, and Proposition 12 show that this relationship arises in the inflation case. Changes in quote intensity, which serves as a proxy for trade volume, generate a positive correlation between trading activity and the size of the speculative premium when inflation varies. This positive correlation matches the empirical patterns observed in historical bubble episodes.

Taken together, the evidence suggests that the central mechanism of the model, the interaction of monetary policy, participation costs, and decentralised exchange, organises several observed features of these markets. The key margins are the broker's decision to post and the intensity with which investors search. Variations in these margins, driven by monetary and regulatory conditions, are sufficient to account for the movements in depth, stability, spreads, and speculative premia documented above.

7 Conclusion

This paper develops a monetary model of financial exchange to examine how monetary and regulatory policy shape price formation, trading activity, and market stability. The framework combines search frictions and costly posting in a unified environment that links the microeconomic process of trade to the macroeconomic role of money. The analysis shows that participation costs and monetary conditions jointly determine the dispersion of bid and ask prices, execution costs, and the probability of market disruption. Increases in posting costs or reductions in real balances make quoting less frequent and prices more dispersed, producing thinner participation and greater variability in execution outcomes. Even small policy adjustments, such as moderate increases in inflation or higher regulatory posting costs, can substantially raise the likelihood of temporary trading breakdowns and flash-crash events. Overall, the results show that financial stability depends not only on aggregate policy settings but also on how these policies influence the incentives governing decentralised trade. Future work could explore how credit conditions and leverage shape the transmission of monetary and regulatory policy to asset prices, particularly through mechanisms involving margin lending and short selling. Such extensions would provide further insight into how the structure of financial intermediation amplifies or dampens the effects of policy on market stability.

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